

Scale-sharp fractional regularity for the low-exponent spherical maximal problem

Candidate partial result for arXiv:1306.6503

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Status. Candidate substantial partial result, likely valid; human review is needed. The full low-exponent $W^{1,p} \rightarrow \dot{W}^{1,p}$ conjecture remains open.

1 Source question

For normalized surface measure σ on S^{n-1} , write

$$\mathcal{S}u(x) := \sup_{r>0} \left| \int_{S^{n-1}} u(x - r\omega) d\sigma(\omega) \right|.$$

The same argument applies to the positive convention with $|u|$ inside the average. Hajlasz and Liu ask whether

$$\mathcal{S} : W^{1,p}(\mathbb{R}^n) \longrightarrow \dot{W}^{1,p}(\mathbb{R}^n) \tag{1.1}$$

is bounded for every $n \geq 2$ and $1 < p < n$ [1]. Stein–Bourgain settles (1.1) for $p > p_c := n/(n-1)$, so the unresolved range is $1 < p \leq p_c$.

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which possibly could exclude it from being in the Sobolev space. However, Theorem [1.3](#) suggests the following question.

Question 1.4. Is the spherical maximal operator bounded in the Sobolev space $\mathcal{S} : W^{1,p}(\mathbb{R}^n) \rightarrow \dot{W}^{1,p}(\mathbb{R}^n)$ for all $n \geq 2$ and $1 < p < n$?

Example [1.5](#) below shows that in the case $p = 1$ the corresponding question has the negative answer.

Figure 1: Question 1.4 in the source paper [1], PDF page 5.

The source proves the pointwise growth estimate $\mathcal{S}u \lesssim I_1(|\nabla u|)$, hence $\mathcal{S}u \in L^{p^*}$ for $p^* = np/(n-p)$, but notes that this does not control oscillation. The next result supplies scale-sharp positive fractional regularity in the entire unresolved range.

2 Strong fractional regularity

Theorem 1. *Let $1 < p < n$, $p^* = np/(n-p)$, and*

$$\max\{p, p_c\} < q < p^*.$$

Set

$$\alpha := 1 - n \left(\frac{1}{p} - \frac{1}{q} \right) \in (0, 1).$$

Then every $u \in W^{1,p}(\mathbb{R}^n)$ satisfies

$$\sup_{h \neq 0} |h|^{-\alpha} \|\tau_h \mathcal{S}u - \mathcal{S}u\|_{L^q} \leq C_{n,p,q} \|\nabla u\|_{L^p}. \quad (2.1)$$

Equivalently, $\mathcal{S}u \in \dot{B}_{q,\infty}^\alpha$ with the displayed homogeneous bound.

Proof. Translation invariance and the triangle inequality for the supremum give the pointwise estimate

$$|\tau_h \mathcal{S}u - \mathcal{S}u| \leq \mathcal{S}(\tau_h u - u). \quad (2.2)$$

Put $g_h = \tau_h u - u$. The fundamental theorem on lines and Sobolev embedding yield

$$\|g_h\|_p \leq |h| \|\nabla u\|_p, \quad \|g_h\|_{p^*} \leq 2 \|u\|_{p^*} \leq C_{n,p} \|\nabla u\|_p. \quad (2.3)$$

Because

$$\frac{1}{q} = \frac{\alpha}{p} + \frac{1-\alpha}{p^*},$$

interpolation of the two norms in (2.3) gives

$$\|g_h\|_q \leq C |h|^\alpha \|\nabla u\|_p. \quad (2.4)$$

Finally, $q > p_c$, so the strong spherical-maximal theorem applied to (2.2)–(2.4) proves (2.1). $\square$

Corollary 2. *Under the hypotheses of Theorem 1, for every $0 < s < \alpha$,*

$$\mathcal{S} : W^{1,p}(\mathbb{R}^n) \longrightarrow W^{s,q}(\mathbb{R}^n)$$

is bounded. In particular, if $p = p_c$, every order $s < 1$ is available with $q > p_c$ sufficiently close to p_c . If $1 < p < p_c$, every order

$$0 < s < n \left(1 - \frac{1}{p} \right) \quad (2.5)$$

is available with a suitable $q > p_c$.

Proof. Since $p \leq q \leq p^*$ and $u \in W^{1,p}$, interpolation gives $u \in L^q$; the strong maximal theorem gives $\mathcal{S}u \in L^q$. In the difference quotient formula for $[\mathcal{S}u]_{W^{s,q}}$, (2.1) controls $|h| < 1$ and the L^q norm controls $|h| \geq 1$. The radial h -integral near zero is $\int_0^1 r^{q(\alpha-s)-1} dr < \infty$. $\square$

3 Critical Lorentz refinement

Theorem 3. *Suppose $n \geq 3$ and $1 < p < p_c$. Put*

$$\beta := n \left(1 - \frac{1}{p} \right) \in (0, 1).$$

Then

$$\sup_{h \neq 0} |h|^{-\beta} \|\tau_h \mathcal{S}u - \mathcal{S}u\|_{L^{p_c, \infty}} \leq C_{n,p} \|\nabla u\|_{L^p}. \quad (3.1)$$

Proof. Bourgain's endpoint estimate in dimensions $n \geq 3$ is

$$\mathcal{S} : L^{p_c, 1} \longrightarrow L^{p_c, \infty}. \quad (3.2)$$

Real interpolation of (2.3), using $(L^p, L^{p^*})_{\theta, 1} = L^{p_c, 1}$, gives

$$\|g_h\|_{L^{p_c, 1}} \leq C |h|^\beta \|\nabla u\|_p; \quad (3.3)$$

indeed the weight of the order-one L^p estimate is $1 - n(1/p - 1/p_c) = \beta$. Combine (2.2), (3.2), and (3.3). $\square$

The full endpoint Lorentz estimate for the circular maximal operator is false, so Theorem 3 is correctly restricted to $n \geq 3$; see [3] and its discussion of the endpoint obstruction.

4 Sharpness and why this is not yet one derivative

For $u_\lambda(x) = u(\lambda x)$, the right side of (2.1) scales as $\lambda^{1-n/p}$. A $\dot{B}_{q, \infty}^\gamma$ seminorm scales as $\lambda^{\gamma-n/q}$. Hence any homogeneous estimate of this form forces

$$\gamma - \frac{n}{q} = 1 - \frac{n}{p},$$

which is exactly $\gamma = \alpha$. Thus Theorem 1 loses no smoothness beyond the integrability tradeoff needed to enter the known spherical-maximal range.

That tradeoff is nevertheless decisive. On the critical line, $W^{1,p}$ embeds into $B_{q, \infty}^\alpha$; the reverse embedding needed to deduce (1.1) is false. In frequency language, the Besov conclusion has only dyadic ℓ^∞ control, while a full Sobolev derivative requires additional scale summability.

5 Eight upgrade attempts and remaining obstruction

Eight focused routes were audited. Direct L^p difference quotients are circular below p_c . Interpolation before applying the maximal theorem is the successful proof above. Endpoint real interpolation gives Theorem 3, but not a strong derivative. Critical Besov embedding runs in the wrong direction. In the companion maximal-potential representation, differentiating a truncation leaves a boundary spherical average, precisely the unavailable low-exponent term. Stationarity of a maximizing radius cancels only the scalar radial mean, not its first spherical moment. The source's logarithmic radial counterexample at $p = 1$ does not extend to $p > 1$: the tangent-sphere singularity becomes integrable, and the radial vector field has an additional odd cancellation. Finally, an upgrade via vector-valued or variation estimates would need scale summability not supplied by known scalar strong or endpoint bounds.

The associated attempt note records these calculations in detail. No full proof or counterexample survived the audit. The new result should therefore be retained as a substantial partial answer, not labeled a solution of Question 1.4.

6 Verification and literature status

The proof uses only: the line-translation estimate, homogeneous Sobolev embedding, elementary interpolation, the classical strong spherical-maximal theorem, and Bourgain’s endpoint Lorentz theorem. The exponent identities were checked directly and by dilation. The argument works for both common absolute-value conventions for $\mathcal{S}$.

Cheap run indexes and bounded searches through 12 August 2026 found no later paper resolving the exact low-exponent Sobolev conjecture. A later Hajlasz–Liu paper still calls it a conjecture, and recent pointwise-potential work recalls the growth estimate rather than a derivative bound. Novelty confidence is moderate because the literature search was bounded; the mathematical argument here is self-contained once the classical maximal theorems are accepted.

References

- [1] P. Hajlasz and Z. Liu, *Sobolev spaces, Lebesgue points and maximal functions*, J. Contemp. Math. Anal. 49 (2014), 67–80; arXiv:1306.6503.
- [2] P. Hajlasz and Z. Liu, *Maximal potentials, maximal singular integrals, and the spherical maximal function*, Proc. Amer. Math. Soc. 142 (2014), 3949–3961; arXiv:1306.6358.
- [3] A. Seeger, T. Tao, and J. Wright, *Endpoint mapping properties of spherical maximal operators*, J. Inst. Math. Jussieu 2 (2003), 109–144; arXiv:math/0205153.